Primary Survey

Manage Airway / Breathing & Circulation as required

Consider ALS Back-Up

- Threats to Airway
- Narcotic Analgesia
- Burns >20% TBSA

Stop the Burning Process

- Cool thermal burns with cool running water for up to 20 minutes

General Burn Care

- Remove Clothing to affected areas
- Apply Burn Gel and Sterile Dressings
- Prevent hypothermia

Consider management of other traumatic injuries

Consider Pain management

Go to Pain CPG

Electrical Burns

12-Lead ECG
And
Continuous 4-Lead ECG monitoring

Arrhythmias should be managed as per appropriate CPG

Thermal Burns

Immediate IV Fluid Administration Guideline for Burns

	Adults	Pediatric
Altered mental Status or **Signs of Shock**	**250-500ml IV Bolus** Repeat Bolus to a maximum of 2000ml	**10-20ml/kg IV Bolus** Repeat Bolus to a maximum of 40ml/kg
Large Burns Adults: >15% TBSA Pediatric: >10% TBSA	**250-500ml**	**10-20ml/kg**

If time is available:
Administer further Isotonic Crystalloids based on the parkland formula

Prevent Hypothermia

Patient Care Goals

1. Ensure airway protection and support oxygenation, especially with inhalation injuries
2. Accurately assess burn severity and classify burn depth and total body surface area (TBSA)
3. Manage pain and prevent hypothermia
4. Initiate timely fluid resuscitation based on burn size and severity
5. Identify and manage life-threatening complications (e.g., airway compromise, shock, electrical injury)
6. Facilitate prompt transport to an appropriate facility with early notification if major burns are suspected

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients of all ages with:
  - Thermal burns (flame, scald, contact)
  - Chemical burns (acid, alkali)
  - Electrical burns (including lightning)
  - Inhalation injury (smoke, steam, chemical)
  - Radiation burns

Exclusion Criteria

- Isolated minor sunburns or abrasions without functional or cosmetic risk
- Chronic skin ulcers unrelated to acute burn injury

Key Patient Assessment Points

1. **Airway Assessment**
   - Assess airway as per Universal Care CPG.
   - Look for facial burns, singed nasal hairs, carbonaceous sputum, stridor, hoarseness (suggest inhalation injury)
2. **Breathing Assessment**
   - Listen for voice changes, wheezing, or stridor
   - Consider early airway protection in severe facial or inhalation burns
3. **Assess for carbon monoxide (CO) exposure**
   - Suspect CO poisoning in all patients exposed to smoke from enclosed-space fires
   - Clinical signs may include: headache, nausea, vomiting, dizziness, confusion, altered mental status, or loss of consciousness
   - Pulse oximetry may appear normal and is unreliable in CO poisoning
   - History of multiple people affected in the same location is also suggestive
4. **Circulation**
   - Check for signs of hypovolemia, especially in large burns or electrical injuries
   - Evaluate peripheral pulses in circumferential limb burns (risk of compartment syndrome)
5. **Neurological Status**
   - Assess for altered mental status (suggests hypoxia, CO poisoning, or associated trauma)
6. **Burn Depth Classification Assessment**
   - Superficial: red, dry, painful
   - Partial thickness: blistered, moist, very painful
   - Full thickness: leathery, dry, insensate
7. **Burn Size Total Body Surface Area (TBSA) Assessment**
   - Use Rule of Nines for adults
   - Use Lund & Browder chart in children
   - Exclude superficial (1st-degree) burns from TBSA calculation
8. **Associated Trauma or Electrical Injury Assessment**
   - Suspect occult injuries, fractures, arrhythmias
   - Consider prolonged cardiac monitoring in high-voltage cases
9. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

- Red Priority:
  - Airway compromise
  - Burns >25% TBSA
  - Circumferential burns

Patient Management & Interventions

1. **Consider ALS Backup**
   - Request ALS back-up as per Universal Care CPG.
     - ALS should be routinely requested for:
       - Threats to airway.
       - Major burns >20% TBSA.
       - The need for narcotic analgesia.
       - Electrical injuries with arrhythmias.
2. **Airway Management**
   - If airway involvement suspected, request ALS support early for possible intubation is essential
   - Do not delay transport for definitive airway control.
3. **Oxygen Therapy**
   - Administer 100% oxygen via non-rebreather mask to all patients with suspected inhalation injury or carbon monoxide exposure
   - Do not rely on SpO_2 readings alone, as they may be falsely reassuring in CO poisoning
   - High-flow oxygen reduces the half-life of carboxyhemoglobin and improves tissue oxygenation
   - Continue oxygen administration throughout transport and notify receiving hospital of suspected CO exposure
4. **Pain Management**
   - Assess and manage pain as per Pain Management CPG.
5. **Burn Cooling (First Aid)**
   - Cool thermal burns with cool running water for up to 20 minutes if within 3 hours of injury
   - Avoid ice, prolonged cooling, or soaking (may cause hypothermia)
   - Cover burns with clean, non-adherent dressing or plastic wrap
6. **Chemical Burns**
   - Remove clothing and brush off dry chemicals
   - Irrigate continuously with copious water for at least 15–30 minutes
7. **Electrical Burns**
   - Monitor for cardiac arrhythmias
   - Assume deep tissue injury regardless of surface appearance
   - Treat for associated trauma (fall, blast)
8. **Establish IV Access**
   - Indicated for fluid or medication administration.
   - Consider IO access if IV cannot be attained.
9. **Fluid Resuscitation**
   - Indicated for burns >10% TBSA in children or >15% in adults
   - Start fluids prehospital if transport time >30 min or patient is hypotensive
     - Dose:
       - Adults: 250-500ml
       - Pediatric: 10-20ml/kg
   - If time permits:
     - Use Parkland Formula:
       - 4 mL × body weight (kg) × % TBSA
       - Give first half over 8 hours, remainder over next 16 hours
   - Note: only partial thickness and full thickness burns (2nd and 3rd degree) are included in TBSA% calculation.
10. **Prevent Hypothermia**
    - Cover patient with foil blanket or clean sheets

- Keep ambient temperature warm

11. Transport

- Prioritize rapid transport in Red and Yellow priority cases
- Notify hospital early if airway or major burns present

12. Patient Reassessment

- Frequently Reassess the patient as per Universal Care CPG.
- Specific Patient Reassessment Points:
  - Pain score
  - SpO_2 and respiratory rate
  - Mental status
  - Fluid response if initiated
  - Burn progression or signs of infection

Patient Safety Considerations

- Avoid ice packs or creams in the field
- Use only sterile or clean coverings — no cotton
- Do not break blisters or debride
- Avoid overhydration — titrate fluids carefully
- Always consider airway risk even in the absence of early symptoms

Clinical Insights & Additional Guidance

- Be alert for signs of airway compromise even in patients who initially appear stable
- Burns in children, elderly, or involving hands/face/genitals require urgent evaluation and transport.
- Electrical burns may have minimal surface damage with significant internal injury
- Chemical burns may continue tissue damage until thoroughly irrigated
- Pediatric doses and fluid calculations can be guided using Broselow tape
- When in doubt, consult TeleEMS and request ALS backup

Documentation

- Identification: Name, age, gender, weight
- Medical complaint: Burn type and cause (flame, scald, chemical, etc.)
- Injuries/Information: TBSA, depth, location, inhalation injury, associated trauma
- Signs: Vital signs, airway status, pain scale, SpO_2
- Treatment: Oxygen, analgesia, fluids, dressings, decontamination
- Allergies: Drug or material-related
- Medications: Prehospital medications given
- Background: Medical history (e.g., asthma, epilepsy, diabetes)
- Other: Response to treatment, need for ALS, handover information

Key Performance Indicators

1. Assess and document burn TBSA for all burn patients.
2. Administer appropriate analgesia within 10minutes of arrival at patient.

References

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